B. Kalindrome Array

time limit per test 1 second
memory limit per test 256 megabytes
input standard input
output standard output

Rating: 1100 | Tags: greedy, two pointers | https://codeforces.com/problemset/problem/1610/B

An array $[b_1, b_2, \ldots, b_m]$ is a palindrome, if $b_i = b_{m+1-i}$ for each i from 1 to m. Empty array is also a palindrome.

An array is called **kalindrome**, if the following condition holds:

- It's possible to select some integer x and delete some of the elements of the array equal to x, so that the remaining array (after gluing together the remaining parts) is a palindrome.

Note that you don't have to delete all elements equal to x, and you don't have to delete at least one element equal to x.

For example :

- [1, 2, 1] is kalindrome because you can simply not delete a single element.
- [3, 1, 2, 3, 1] is kalindrome because you can choose $x = 3$ and delete both elements equal to 3, obtaining array [1, 2, 1], which is a palindrome.
- [1, 2, 3] is not kalindrome.

You are given an array $[a_1, a_2, \ldots, a_n]$. Determine if a is kalindrome or not.

Input

The first line contains a single integer t ($1 \le t \le 10^4$) — the number of test cases. The description of the test cases follows.

The first line of each test case contains a single integer n ($1 \le n \le 2 \cdot 10^5$) — the length of the array.

The second line of each test case contains n integers $a_1, a_2, \ldots, a_n$ ($1 \le a_i \le n$) — elements of the array.

It's guaranteed that the sum of n over all test cases won't exceed $2 \cdot 10^5$.

Output

For each test case, print `YES` if a is kalindrome and `NO` otherwise. You can print each letter in any case.

Example

Input
4 1 1 2 1 2 3 1 2 3 5 1 4 4 1 4

Output
YES YES NO YES

Note

In the first test case, array [1] is already a palindrome, so it's a kalindrome as well.

In the second test case, we can choose $x = 2$, delete the second element, and obtain array [1], which is a palindrome.

In the third test case, it's impossible to obtain a palindrome.

In the fourth test case, you can choose $x = 4$ and delete the fifth element, obtaining [1, 4, 4, 1]. You also can choose $x = 1$, delete the first and the fourth elements, and obtain [4, 4, 4].